

Every Number in the Grid

An Optional Deeper Pass, With a Project to Match

Optional Reading, Session 05: Building the Grid · Portfolio as Narrative

I. Purpose

The reading assigned for tonight's session establishes the twelve-point system in outline: a governing unit, a page built as a multiple of it, and margins chosen the same way. It does not walk through every number that follows from those choices, nor does it ask a student to do anything with the arithmetic beyond confirming it once. This reading exists for students who want both of those things: a complete account of where every number in the grid comes from, and an assignment that puts the arithmetic to use inside a real project rather than a blank test file. Neither the reading nor the assignment it describes is required. Both are available to anyone who wants a second pass at the system before moving on.

II. The Unit and the Page

Twelve points is the unit this class works from. Every other number in the grid, without exception, is derived from it, not chosen independently. The document itself, six hundred by eight hundred forty points, is the first derivation: eight hundred forty divided by twelve equals seventy exactly, meaning the full page height is seventy discrete twelve-point increments, end to end, before a single margin or guide is drawn. Six hundred divided by twelve equals fifty, confirming the width resolves the same way.

III. Margins, and Why They Are Multiples of Twelve

The margins, thirty-six points on the top, inside, and outside edges and forty-eight on the bottom, are three and four multiples of twelve respectively. This is not a coincidence of clean numbers. It is a requirement. Subtracting the margins from the page produces the content area, and that subtraction has to leave a result that is itself still a multiple of twelve, or every measurement built on top of it, rows, columns, baseline lines, inherits the fraction and never resolves cleanly. A margin of forty points, for instance, is not smaller or larger in any way that matters visually. It is simply not a multiple of twelve, and choosing it would corrupt every calculation that follows for the rest of the file.

IV. The Row Math, Traced Completely

Subtract the top and bottom margins from the page height: eight hundred forty minus thirty-six minus forty-eight equals seven hundred fifty-six points of content height. Divide that by twelve:

seven hundred fifty-six divided by twelve equals sixty-three exactly. Call each of those twelve-point increments a baseline unit, the smallest measured step this file recognizes vertically. The content area is sixty-three baseline units tall, no remainder, which is the condition that makes everything past this point possible.

Those sixty-three baseline units now have to be distributed across eight rows. Each row consumes seven baseline units of visible content, and each of the seven gaps between rows consumes one baseline unit as a gutter. Eight rows of seven baseline units is fifty-six. Seven gutters of one baseline unit is seven. Fifty-six plus seven is sixty-three, matching the content height exactly, with nothing left over and nothing borrowed. In points, each row is seven baseline units times twelve points, eighty-four points tall, and each gutter is twelve points, one baseline unit exactly. Because every row and every gutter is a whole number of baseline units, every boundary between them lands on an actual baseline line. This is the property that makes the grid feel inevitable rather than approximate once it is built: nothing in the row structure was rounded to get there.

V. The Column Math, Traced Completely

Columns follow a different derivation, and a different rule. Subtract the inside and outside margins from the page width: six hundred minus thirty-six minus thirty-six equals five hundred twenty-eight points of content width. Six columns with five twelve-point gutters between them consume sixty points of that width in gutters alone; five hundred twenty-eight minus sixty equals four hundred sixty-eight, and four hundred sixty-eight divided by six equals seventy-eight points per column, exactly, no remainder.

Seventy-eight points is not a whole number of baseline units. Seventy-eight divided by twelve is six and one half. This is not an error, and it does not need to be fixed. Columns are vertical dividers; the baseline grid is a set of horizontal lines. The two never intersect in a way that requires agreement, because a column's width has no bearing on where a line of text sits vertically. Only measurements that run in the same direction as the baseline grid, margins, row heights, gutters measured top to bottom, are required to be whole baseline units. Column widths are required only to divide the content width evenly among themselves, which seventy-eight points does.

VI. What “Aligned” Actually Means

A grid being present in a file and a grid actually governing that file's content are different conditions, and the difference is exactly what the optional assignment below tests. An element is aligned when two things are both true: its edges, read off the Control panel as X, Y, width, and height, land on the running sequence of column and row boundaries rather than nearby approximations, and any text it contains has Align to Grid set and visibly locks to a baseline rule

rather than floating between two of them. Neither condition is visible from a glance at the page. Both are checked by selecting an element, reading its coordinates, and comparing those coordinates against the reference tables below, one for column edges, measured from the inside margin, one for row edges, measured from the top margin.

Column left edges (from the inside margin, in points)

1	2	3	4	5	6
0	90	180	270	360	450

Row top edges (from the top margin, in points)

1	2	3	4	5	6	7	8
0	96	192	288	384	480	576	672

An element whose edge reads eighty-eight or ninety-three is not aligned. It is close, and close is the condition this whole system was built to eliminate.

VII. The Optional Assignment: Align Your Actual Project

If you want to go further than tonight requires, open the same project file you have been building all term and do to it what the checklist only asked you to prove you could do on a blank page: move its actual content until every image and text frame resolves to the grid. Concretely:

Select every image and text frame already placed in your file. For each one, read its X, Y, width, and height in the Control panel. Compare each of those four numbers against the column and row boundaries in the tables above. Where an edge is close but not exact, move or resize the frame until it lands on the boundary precisely, not approximately.

Where a frame's width or height does not match a whole number of modules, either resize it to span exactly one, two, or more modules cleanly, or accept that it deliberately breaks the grid, a legitimate choice later in the term, but not one to make by accident this early.

Confirm every paragraph of body text still reads Align to Grid: All Lines, and check that the text visibly locks to a baseline rule rather than sitting between two of them, the same visual check tonight's checklist asked for, now run against real content instead of a blank frame.

This is not a small task, and it is not expected to be finished in one sitting. It is the actual work of laying out a portfolio page, done once, early, so that every session from here forward starts from a file that already behaves the way it is supposed to.

VIII. If You Are Printing

The system in this reading and in tonight's class uses six hundred by eight hundred forty points because the arithmetic is the cleanest version to learn on. It is not a real sheet of paper. If your final portfolio will be printed, the size to use is six hundred twelve by seven hundred ninety-two points, true Letter, eight and a half by eleven inches. That page also divides evenly by twelve, seven hundred ninety-two divided by twelve is sixty-six, but its rows do not divide as cleanly under the same margins used here: content height becomes seven hundred eight points, and eight rows of that height come out to seventy-eight points each, six and one half baseline units, the same kind of fractional result columns are allowed to have but rows are not. Fixing it means enlarging the bottom margin to ninety-six points, which restores whole seventy-two point rows, six baseline units each, at the cost of a visibly larger bottom margin than the six hundred by eight hundred forty version uses. Neither version is wrong. They trade a real paper size against a slightly different margin proportion, and the choice belongs to whoever is printing, not to the arithmetic.

IX. Conclusion

Every number examined in this reading, the page, the margins, the rows, the columns, the module boundaries a real image or text frame is checked against, comes from the same twelve-point unit and no other source. None of them were chosen for their own sake. They were derived, in sequence, each one constrained by the one before it, and the optional assignment this reading describes is simply the practice of holding a real project to that same derivation instead of trusting it by inspection. A grid that only exists in the abstract, present in the file but never checked against actual content, is not yet doing its job. This is the reading for students who want to see it do that job once, deliberately, before the rest of the term asks them to do it every week.

References

Portfolio as Narrative, Session 05: Building the Grid, lecture and in-class activity.

“The Inch Was Already Twelve,” Session 05 required reading, Portfolio as Narrative.